

Calculus (PR12)

Week 3: Optimization, Taylor and Maclaurin Series

Learning Goals



Critical points and optimization

- Identify **maxima and minima** using derivatives
- Connect optimization to **AI training**



Concavity by derivatives

- Use the **second derivative** to analyze concavity
- Recognize **inflection points** and change in behavior



Local approximation

- Use **linear** and **Taylor approximations**
- Interpret Taylor polynomials as **local predictions** of functions



Connection to AI

How does Spotify predict your next song? How does ChatGPT improve its answers? How does Google Maps optimize routes?

All of these rely on:

- 1. Finding the “best value”**
- 2. Understanding how things change**
- 3. Approximating complicated functions**

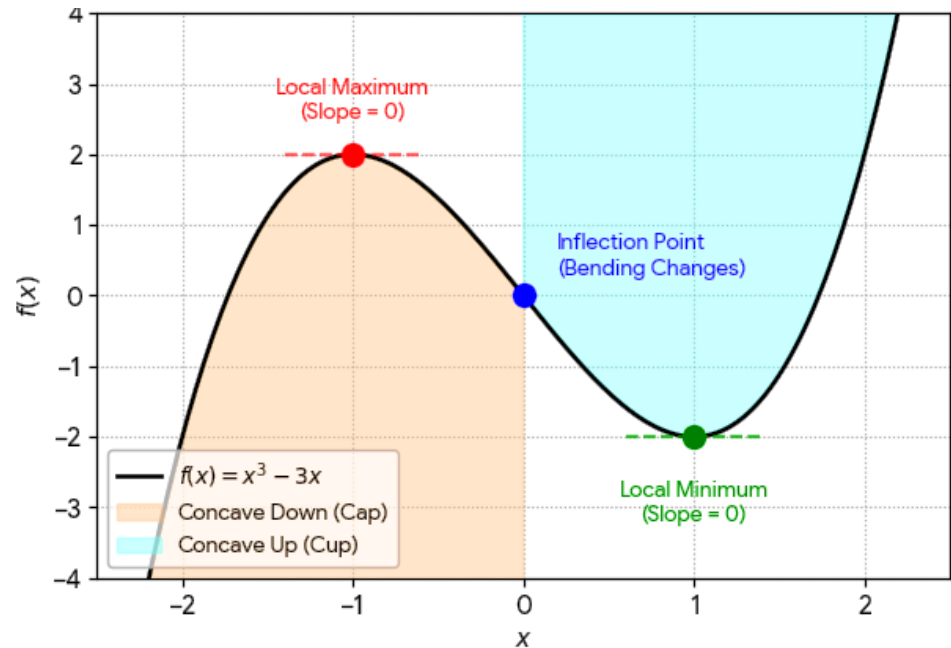
Today's big idea: Calculus helps AI make smart local decisions.

Special Points

Question: Which point is "special"?

**Consider a hill while cycling.
At which point are you:**

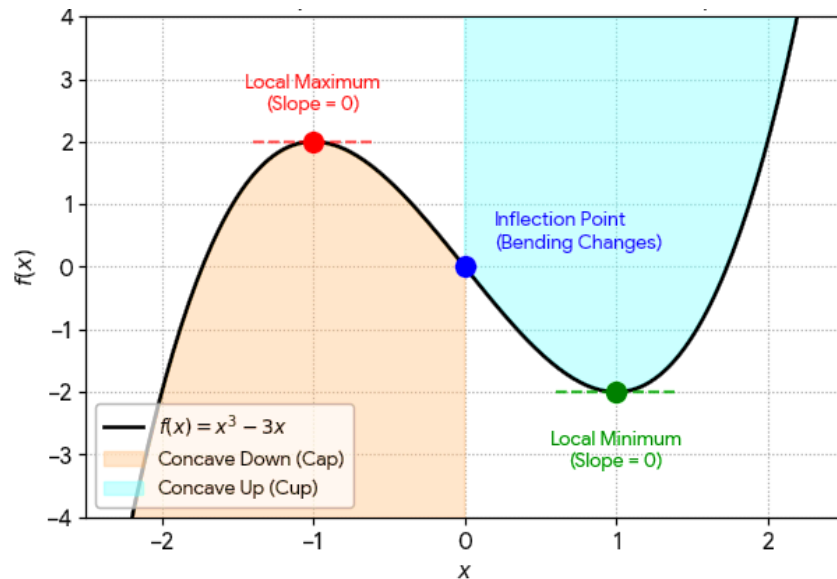
- Moving uphill fastest?
- Neither going up or down?
- Changing from "cup-shaped" to "cap-shaped"?



Critical Points = Candidate Best/Worst Locations

Intuition: critical points are (important) places where:

- The slope becomes zero, OR
- The derivative does not exist



Examples of daily life:

- Lowest fuel consumption
- Highest profit
- Fastest learning rate in AI training
- Best exam score prediction

Setting Derivative Equal to Zero

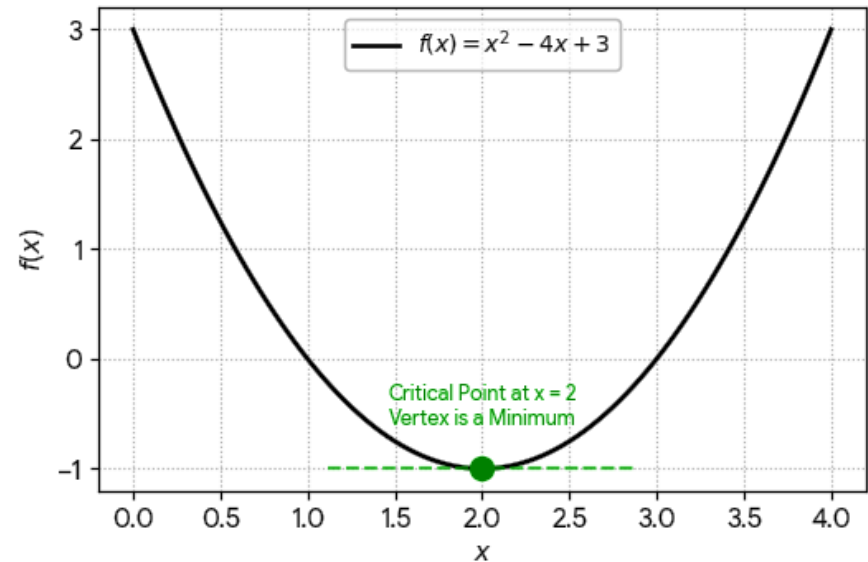
Example:

$$f(x) = x^2 - 4x + 3$$

Derivative:

$$f'(x) = 2x - 4$$

$$f'(x) = 0 \Rightarrow 2x - 4 = 0 \Rightarrow x = 2$$



Question: Is this point a maximum or minimum?

First Derivative Test

Think like a hiker: if the slope (sign) changes:

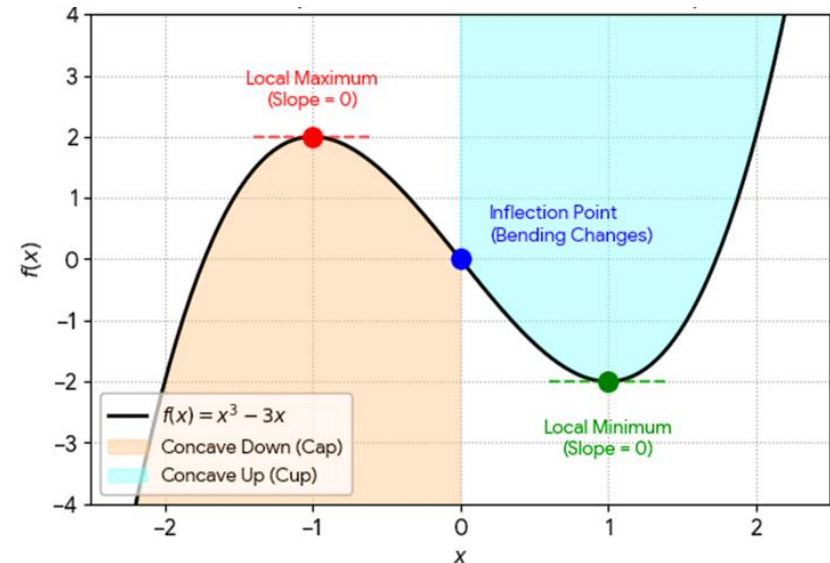
- From positive to negative $\rightarrow$ maximum
- From negative to positive $\rightarrow$ minimum

Before Points	After Point	Meaning
+	-	Maximum
-	+	Minimum

Question: Gradient descent in ML follow a minima or maxima?
Why?

Second Derivative as Shape Detector

- **First Derivative** tells direction.
- **Second derivative** tells shape.



Concavity:

$f''(x) > 0$: bowl/cup shape $\rightarrow$ concave **UP**

$f''(x) < 0$: upside-down bowl $\rightarrow$ concave **DOWN**

Example

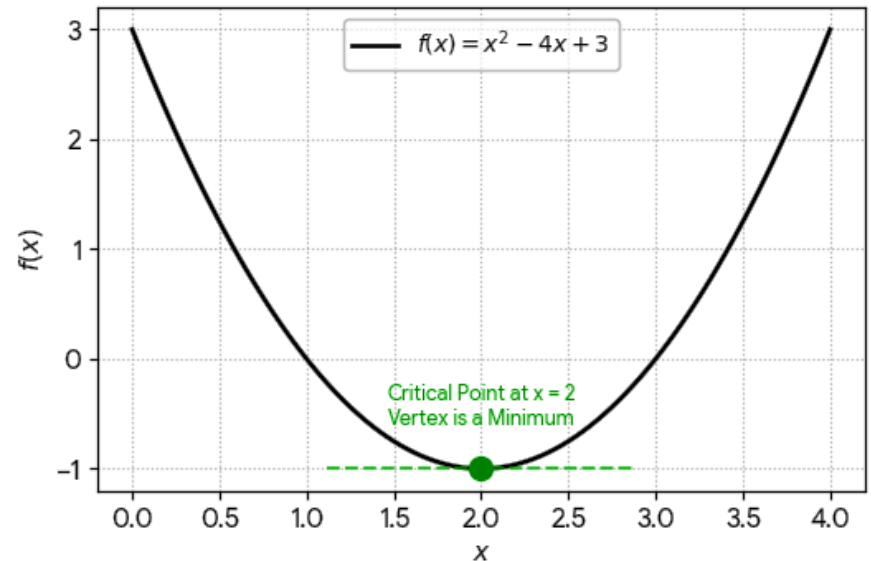
$$f(x) = x^2 - 4x + 3$$

First derivative:

$$f'(x) = 2x - 4$$

second derivative:

$$f''(x) = 2 > 0$$



Question: Is this point a maximum or minimum?

Inflection Points

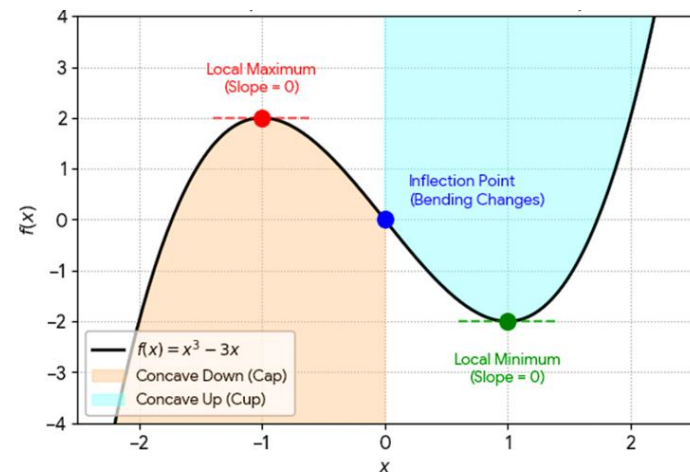
Definition: An inflection point is where the graph changes bending direction.

Example:

$$f(x) = x^3 - 3x$$

$$f'(x) = 3x^2 - 3$$

$$f''(x) = 6x$$



At $x = 0$:

- Second derivative changes sign
- Curvature changes

From Exact to Approximation

Important question:

Do we always need the full complicated function?

Usually: NO.

Linear Approximation = Zooming In

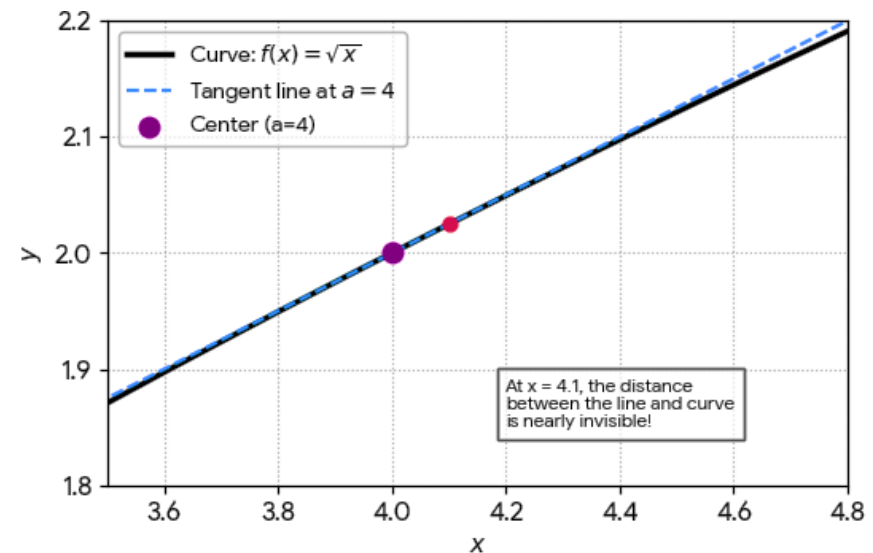
Main intuition: If you zoom in enough, most smooth curves look like straight lines.

Tangent line approximation:

$$f(x) \approx f(a) + f'(a)(x - a)$$

Question:

Why is choosing a nearby point important?



Example Linear Approximation

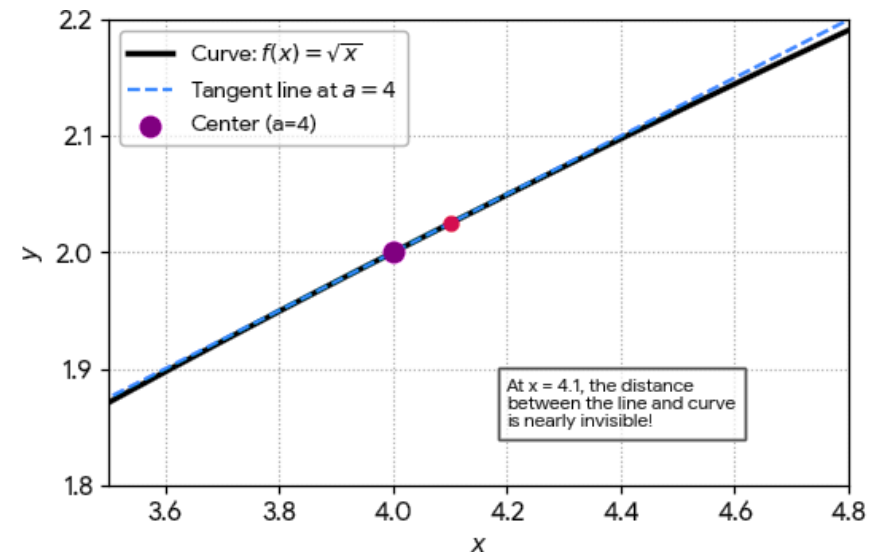
Example: approximate $\sqrt{4.1}$ by using $f(x) = \sqrt{x}$ near $a = 4$:

We know $f(4) = 2$ and:

$$f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(4) = \frac{1}{4}$$

Approximation:

$$\sqrt{4.1} \approx 2 + \frac{1}{4} (0.1) = 2.025$$



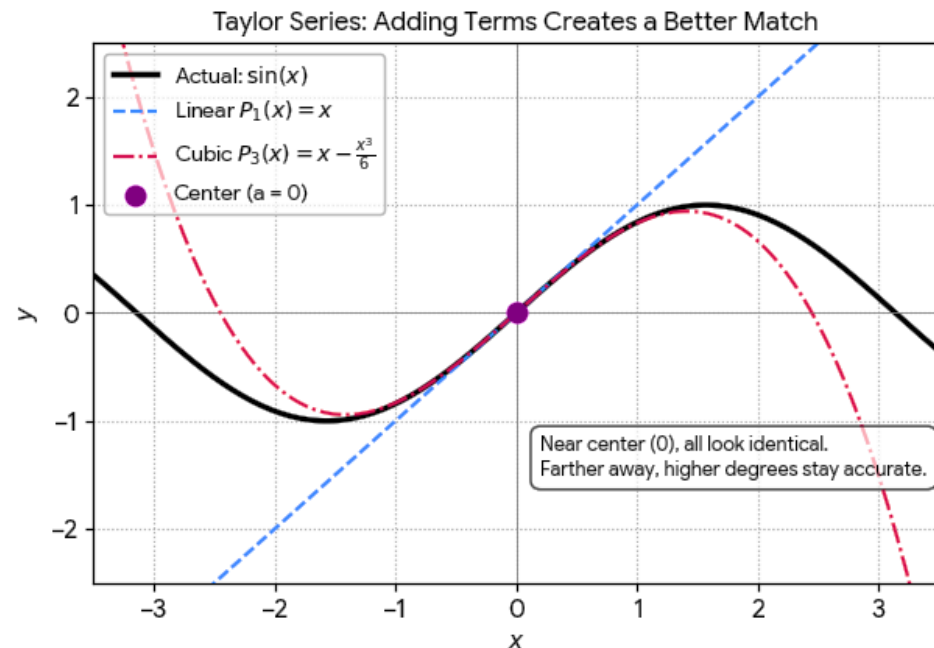
Taylor Polynomials: Better Predictions

Linear approximation uses:

- **Value**
- **Slope**

But curves also bend. Taylor polynomials add:

- **Curvature**
- **Higher-order behavior**
- **More local information**



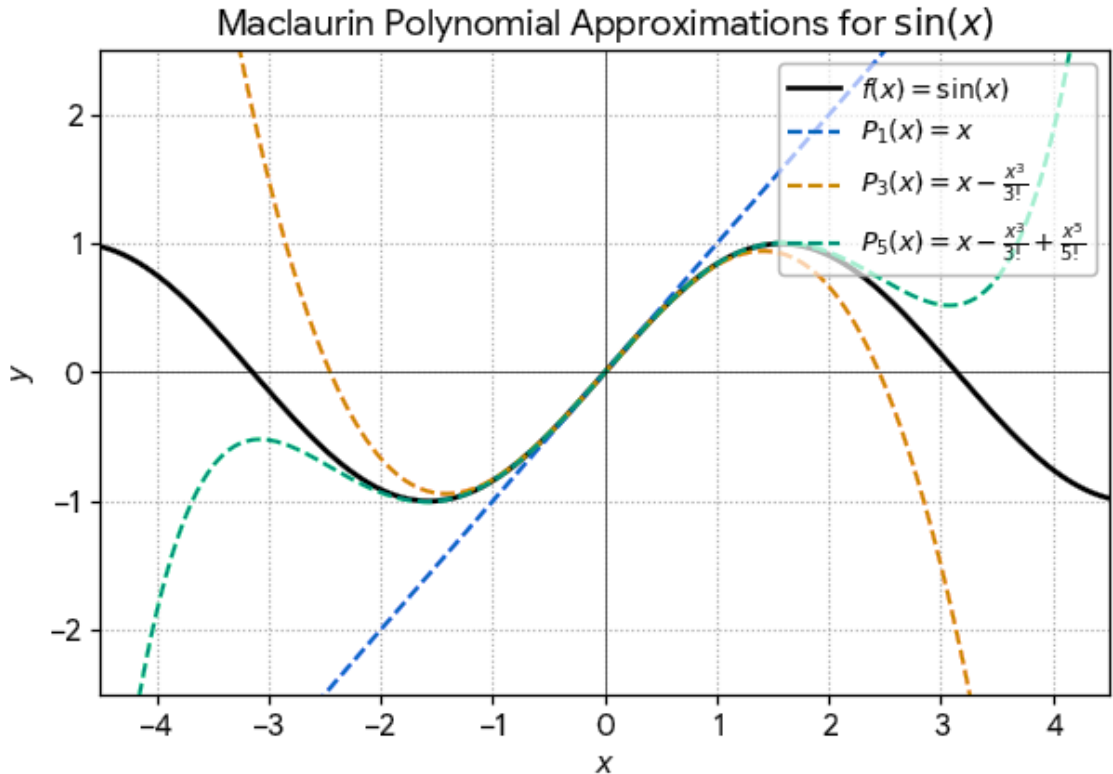
Taylor Polynomial of n -th Degree

$$\begin{aligned} P_n(x) &= f(a) + f'(a)(x - a) \\ &+ \frac{f''(a)}{2!} (x - a)^2 + \frac{f'''(a)}{3!} (x - a)^3 + \dots \\ &+ \frac{f^{(n)}(a)}{n!} (x - a)^n \end{aligned}$$

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k$$

Maclaurin Polynomial of n -th Degree

Each Extra term improves **local accuracy**.



$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} (x - 0)^k$$



Maclaurin Polynomial of $f(x)$

$$f(x) = \frac{1}{1-x}$$

$$f'(x) = -1(1-x)^{-2} \cdot (-1) = (1-x)^{-2}$$

$$f''(x) = -2(1-x)^{-3} \cdot (-1) = 2(1-x)^{-3}$$

$$f'''(x) = -3 \cdot 2(1-x)^{-4} \cdot (-1) = 6(1-x)^{-4}$$

$$f^{(k)}(x) = k! (1-x)^{-(k+1)}$$

Maclaurin Polynomial of $f(x)$

$$f(0) = \frac{1}{1-0} = 1$$

$$f'(0) = (1-0)^{-2} = 1$$

$$f''(0) = 2(1-0)^{-3} = 2$$

$$f'''(0) = 6(1-0)^{-4} = 6$$

$$f^{(k)}(0) = k! (1-0)^{-(k+1)} = k!$$

Apply the Taylor Formula

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k$$

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(k)}(0)}{k!}x^k$$

$$P_n(x) = 1 + \binom{1}{1}x + \frac{2}{2!}x^2 + \frac{6}{3!}x^3 + \dots + \frac{k!}{k!}x^k$$

$$P_n(x) = 1 + x + x^2 + x^3 + \dots + x^k$$

Mini Exercise - Interpreting Derivative Geometrically

Suppose:

$$f'(2) = 0$$

and:

$$f''(2) < 0$$

Question: What can we conclude about the graph near $x = 2$?

The Relevance to Computers and AI

Computers often approximate expensive functions using series.

Examples:

- **GPUs**
- **Scientific computing**
- **Neural network libraries**
- **Physics engines**

Question: Is approximation better than exact computations for interactive prompts?



Big Picture Connection to AI

Today's concepts appear everywhere in AI.

Calculus Concept	AI Application
Critical points	Optimizing loss function
Derivatives	Gradient descent
Concavity	Optimization behavior
Taylor approximation	Fast computation
Local approximation	Neural network updates



Check?

1. Why are critical points important?
2. What does the second derivative tell us?
3. Why are Taylor polynomials useful?
4. Where might approximation appear in AI systems?

Remember:

- $f'(x) = 0$ does NOT always mean maximum/minimum, it needs a sign change or second derivative test.
- Taylor polynomials work best near the center point, approximations become worse farther away

Next week:

Indefinite integration techniques.



?

Questions

- **Follow-up reading: Calculus: A Complete Course, (sections 4.4 – 4.5 & 4.9 – 4.10 & 9.6)**
- **Joining the Discussion Groups and the Lab**

